

CONTACT DETAILS	Email: ghoshadi@stanford.edu Address: Computing and Data Science (CoDa) 389 Jane Stanford Way, Stanford University Stanford, CA 94305 Webpage: ghoshadi.github.io
EDUCATION	Ph.D. in Statistics, Stanford University GPA 4.13/4.00 2022 – 2027 (expected) • Advised by Stefan Wager (Stanford GSB) & Dominik Rothenhäusler (Stanford Statistics) Masters of Statistics (M.Stat), Indian Statistical Institute , Kolkata 2020 – 2022 • Dissertation advisor: Prof. Bodhisattva Sen (Columbia University) Bachelor of Statistics (B.Stat), Indian Statistical Institute , Kolkata 2017 – 2020
RESEARCH	My current research focuses on designing robust methods for causal inference in complex settings that arise in modern applications, such as dynamic systems that evolve over time, units interacting with shared resources, or heterogeneity across different locations and time periods. I like to build methods that deal with these complexities while being as rigorous and transparent as the more classical causal inference methods. Ghosh, A. & Wager, S. (2025+). Non-parametric Causal Inference in Dynamic Thresholding Designs. <i>Major revision at the Journal of the Royal Statistical Society Series B: Statistical Methodology</i>; extended abstract accepted at ACM EC 2026 ↗ arXiv ↗ Summary: Threshold-based interventions are common in clinical practice, but studying their long-term impact is challenging due to temporal dynamics and carryover effects. We show that a dynamic marginal policy effect at the treatment threshold admits a simple, reduced-form characterization, and develop a local linear regression method for estimation and inference. Ghosh, A. & Rothenhäusler, D. (2025+). Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies. <i>Major revision at the Annals of Statistics</i>. arXiv ↗ R package ↗ Summary: Covariate adjustment is essential for removing confounding in observational studies, yet researchers often face uncertainty over which covariates to adjust for. Building on debiased machine learning, we establish valid inference for a reweighted target population without committing to a single specification—requiring only that at least one candidate adjustment is valid. Ghosh, A., Imbens, G. & Wager, S. (2025+). PLRD: Partially Linear Regression Discontinuity Inference. <i>Submitted</i>. arXiv ↗ R package ↗ Summary: Regression discontinuity designs are widely used in applied economics, yet the existing methods often face a trade-off between nominal coverage and interval width. Building on the bias-aware literature, PLRD exploits a partially linear structure and higher-order smoothness to produce intervals that are both valid and short. Ghosh, A., Deb, N., Karmakar, B., & Sen, B. (2026). Robustness and Efficiency of Rosenbaum’s Rank-based Estimator in Randomized Trials: A Design-based Perspective. <i>Accepted at Biometrika</i>. arXiv ↗ Journal link ↗ Summary: How to draw design-based causal inference using rank-based methods from nonparametric statistics? We study robustness and efficiency properties of a rank-based estimator proposed by Rosenbaum (1993) under the design-based framework.

1. **Ghosh, A.** (2019). An asymptotic formula for the Chebyshev theta function. *Notes on Number Theory and Discrete Mathematics*, 25(4), 1-7. [Journal link](#) 
[Summary: Derived sharp bounds on the geometric mean of the first \$n\$ prime numbers.](#)

ACADEMIC SERVICES

- I help organize the **Online Causal Inference Seminar**, a weekly seminar on causal inference with participants from all over the world.
- Served as a reviewer for: *AoS*, *JASA*, *J. Econom.*, *JRSSB*, *NeurIPS*

TEACHING

- As instructor, Stanford University** [ExploreCourses](#) 
- Stats 302: Qualifying Exam Workshop (Theoretical Statistics). Summer 2025, 2026
 - Stats 302: Qualifying Exam Workshop (Probability). Summer 2024
- As teaching assistant, Stanford University** [ExploreCourses](#) 
- Stats 361: Causal Inference. Winter 2025
 - Stats 310B/Math 230B: Theory of Probability II. Winter 2024
 - Stats 310A/Math 230A: Theory of Probability I. Autumn 2023
 - Stats 216: Introduction to Statistical Learning. Winter 2023
 - Stats 209: Introduction to Causal Inference. Fall 2025
 - Stats 203: Regression Models and Analysis of Variance. Fall 2026
 - Stats 202: Data Mining and Analysis. Summer 2023, Autumn 2022
 - Stats 200: Introduction to Theoretical Statistics. Winter 2026, Autumn 2024
 - Stats 141: Introduction to Statistics for Biology. Spring 2026
 - Stats 60: Introduction to Statistical Methods: Precalculus. Spring 2025

Other experiences

- Trained numerous high school students for mathematics olympiads and other math competitions; materials are available at ghoshadi.wordpress.com .

ACCOLADES

- **Departmental Teaching Assistant Award**, Stanford University 2026
- **IMS 2025 International Conference on Statistics and Data Science (ICSDS)**, student travel award 2025
- **IISA conference 2025**, best poster award 2025
- **Industrial Affiliates Annual Conference at Stanford**, travel award 2024
- Selected for **Machine Learning in Economics Summer Institute** at University of Chicago Booth school of business 2024
- **ISIAA – J. K. Ghosh Memorial Gold Medal** (for outstanding performance in Masters of Statistics) 2023
- **ISIAA – Mrs. M. R. Iyer Memorial Gold Medal** (for best overall performance in Bachelor of Statistics) 2021
- **Nikhilesh Bhattacharyya Memorial Gold Medal** (for best performance in major (Statistics) in Bachelor of Statistics) 2021
- **Madhava Mathematics Competition**, selected among the top few participants in the country for a prestigious summer camp 2019, 2018
- **Indian National Mathematical Olympiad**, earned a prestigious certificate of merit from NBHM, Govt. of India (awarded to the top 75 participants in the country) 2016

INVITED TALKS	• 2026 California Econometrics Conference in Honor of Guido Imbens 2026 Title: TBD. • Women and Allies in Statistics (WAIS) Research Talk Workshop 2026 • Joint Statistical Meeting, Boston, Massachusetts 2026 Title: Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies. • ACM Conference on Economics and Computation (EC'26), Rome, Italy 2026 Title: Nonparametric Causal Inference in Dynamic Thresholding Designs • Metrics-lunch (an weekly seminar on Econometrics), Stanford University 2026 Title: Non-parametric Causal Inference in Dynamic Thresholding Designs • Data-driven Seminar, Stanford University 2026 Title: Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A Design-based Perspective • University of Washington Causal Inference & Missing Data Reading Group 2026 Title: Assumption-robust Causal Inference • IMS International Conference on Statistics and Data Science, Seville, Spain 2025 Title: Assumption-robust Causal Inference • Stanford Causal Science Center (SC²) Conference, Stanford University 2025 Title: Causal Inference in Dynamic Thresholding Designs • Joint Statistical Meeting, Nashville, Tennessee 2025 Title: Causal Inference in Dynamic Thresholding Designs • Industrial Affiliates Annual Conference, Stanford University 2024 Title: Practical Bias-aware Inference in Regression Discontinuity Designs • Stanford Causal Science Center (SC²) Conference, Stanford University 2024 Title: Asymptotic bias-aware inference in regression discontinuity designs under higher-order smoothness • Computational and Methodological Statistics, Berlin, Germany 2023 Title: Efficiency and robustness of Rosenbaum's regression (un)-adjusted rank-based estimator in randomized experiments • PCM Memorial Lecture, Indian Statistical Institute, Kolkata 2022 Title: The synthetic control method in causal inference • D. Basu Memorial Lecture, Indian Statistical Institute, Kolkata 2021 Title: Large low-rank matrix completion
	• Online Reading Group on Functional Data Analysis ↗ 2021 Title: Two-sample testing of the equality of mean functions • Students' Learning Seminar, Indian Statistical Institute, Kolkata 2021 Title: Matching estimators in causal inference
CLASS PROJECTS	• SMARTer Multi-task Fine-tuning of BERT Summary: Engineered a multi-task NLP framework, integrating adversarial regularization, proximal optimization, and low-rank adaptation with randomized task scheduling for stable cross-task training. Joint with Disha Ghandwani and Rahul Kanekar.

- **Inference for Data Adaptively Sampled via REINFORCE**

Summary: Developed inference methods for adaptively collected data and analyzed the robustness–efficiency trade-offs in variance estimation for long-horizon policy evaluation.

Joint with Ivy Zhang.

- **Predicting the Number of Micro-businesses Across the U.S.**

Summary: Built a forecasting pipeline using GoDaddy panel and Census data, applying ARIMA, GLMs, and varying-coefficient models to link demographics with county-level microbusiness dynamics. Joint with Joshua Kazdan and Yash Nair.

SKILLS

Python, **R**, Markdown, Git, C (basic), HTML (basic). **R packages:** `plrd` [↗](#) `specrobust` [↗](#)